

Regression Analysis (I)

Solutions to Final Exam.

June 17, 2019

1.(200) Consider a multiple linear regression model $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$, $\boldsymbol{\epsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$, where $\mathbf{X}$ is $n \times p$ design matrix, and $\boldsymbol{\beta} = (\beta_0, \dots, \beta_{p-1})^T$.

(1)(50) Find the distribution of SSR/σ^2 . (Ans) $\chi^2(p-1; \lambda)$, where $\lambda = \boldsymbol{\beta}^T \mathbf{X}^T (\mathbf{I} - \mathbf{J}/n) \mathbf{X} \boldsymbol{\beta}$, i.e., non-central chi-square dist. (Done in class and refer to the lecture note for details)

(2)(50) Find the distribution of $SSR/(p-1)s^2$, where $s^2 = \sum_{i=1}^n e_i^2/(n-p)$ and e_i is the i th residual. (Ans) Can show $SSE/\sigma^2 \sim \chi(n-p)$. Also, can show SSR and SSE are independent. Therefore, $SSR/(p-1)s^2 \sim F((p-1), (n-p); \lambda)$, i.e., non-central F dist.

(4)(50) Let h_{ij} be the ij -th component of $\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}$. Show that $\sum_{j=1}^n (e_j + h_{ij})^2 - h_{ii} = SSE$. (Ans) $\sum_{i=j}^n (e_j + h_{ij})^2 - h_{ii} = \sum_j e_j^2 + 2 \sum_j e_j h_{ij} + \sum_j h_{ij}^2$. Now, by the idempotency of $\mathbf{H}$ and $\mathbf{X}^T \mathbf{e} = \mathbf{0}$, we have $\sum_j h_{ij}^2 = h_{ii}$ and $\sum_j h_{ij} e_j = 0$. Therefore, $\sum_{j=1}^n (e_j + h_{ij})^2 - h_{ii} = SSE$.

2.(50) Let Y_1, Y_2 be iid $N(0, 1)$, $\mathbf{A} = (a_{ij})$ and $\mathbf{B} = (b_{ij})$ be 2×2 matrices with $a_{ii} = 1, a_{ij} = -1, i \neq j$; $b_{ij} = 1, \forall i, j$. Find the distribution of $\mathbf{y}^T \mathbf{A} \mathbf{y} / \mathbf{y}^T \mathbf{B} \mathbf{y}$, where $\mathbf{y} = (Y_1, Y_2)$. (Ans) Note that $\mathbf{A}/2$ and $\mathbf{B}/2$ are idempotent. Also, $\mathbf{AB} = \mathbf{O}$. Therefore, $\mathbf{y}^T \mathbf{A} \mathbf{y} / 2 \sim \chi^2(1)$, $\mathbf{y}^T \mathbf{B} \mathbf{y} / 2 \sim \chi^2(1)$, and they are independent. Hence, $\mathbf{y}^T \mathbf{A} \mathbf{y} / \mathbf{y}^T \mathbf{B} \mathbf{y} \sim F(1, 1)$.

3.(100) Let $Y_1, \dots, Y_n$ be iid with mean μ and variance σ^2 . Let $Q = \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2 / (2n-2)$.

(1)(50) Express Q in a quadratic form. (Ans) $(2n-2)Q = y_1^2 + 2 \sum_{i=2}^{n-1} y_i^2 - 2 \sum_{i=1}^{n-1} y_i y_{i+1}$. Let $(2n-2)Q = \mathbf{y}^T \mathbf{A} \mathbf{y}$, then $a_{11} = a_{nn} = 1, a_{ii} = 2, i = 2, \dots, n-1$ and $a_{ij} = -1, \forall i \neq j$.

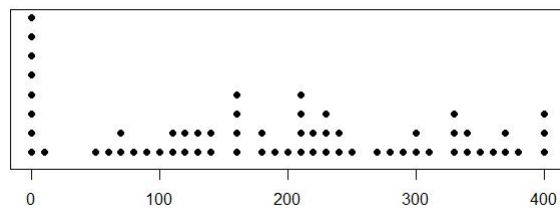
(2)(50) Show that Q is unbiased estimator of σ^2 . (Ans) Recall that $E(\mathbf{y}^T \mathbf{A} \mathbf{y}) = \boldsymbol{\mu}^T \mathbf{A} \boldsymbol{\mu} + \sigma^2 \text{tr}(\mathbf{A})$, then it is easy to show that $E(Q) = \sigma^2$.

4.(50) Consider the AR(1) model : $\epsilon_t = 0.5\epsilon_{t-1} + \delta_t$, where δ_t is iid with

$E(\delta_t) = 0$ and $Var(\delta_t) = 1$. Compute $Var(\epsilon_t)$. (Ans) Since $\epsilon_t = \sum_{j=0}^{\infty} (0.5)^j \delta_{t-j}$ and $Cov(\epsilon_t, \epsilon_{t-l}) = \frac{4}{3}(0.5)^l$ (done in class), we have $Var(\epsilon_t) = \frac{4}{3}$.

Summary Statistics for Final Scores

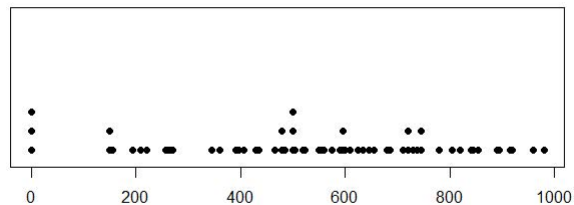
$n = 63$, sample mean = 191



Top Seat : 박혜림, 하도연, 한정욱

Summary Statistics for Total (Midterm + Final + 2 Projects) Scores

$n = 66$, sample mean = 542



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